

Performance of L-band DAS on Live Traffic Fiber Compared to C-band DAS Over Dark Fiber

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Abstract *We present field-trial results with L-band DAS deployed in coexistence with C-band DWDM channels carrying live production traffic. We monitored the quality of DAS and optical communication channels during the test and uncompromised performance was achieved for both compared to sensing on dedicated dark fiber.* ©2026 The Author(s)

Introduction

Research and deployments of distributed acoustic sensing (DAS) have been growing rapidly since 2005 [1]. Developments have focused on technological capabilities as well as investigating the possible exploitation of the sensing data by specific research disciplines, such as seismology [2]. Experimental data acquisition in domain specific research disciplines is typically conducted in the field, and usually DAS has been deployed in dedicated unlit (dark) telecommunications fibers [3]. However, recent developments suggest that DAS signals could coexist with traffic in a live telecommunications cable, with production traffic carried in a separate part of the spectrum. This can be achieved by injecting the DAS signal into the long-band (L-band) area of the optical transmission system [4]. Such a development paves the way for utilizing a greater variety of cables and locations without sacrificing the production channel capacity for telecoms.

In this paper, we present the findings from a field-trial with DAS operating in the L-band of an optical transmission system carrying live data traffic in the conventional band (C-band). Demonstration of DAS and DWDM in coexistence in the same fiber have already been reported [5], but to the best of our knowledge, our work is the first time such a coexistence is validated with live production DWDM channels where optical performance of optical channels have been monitored before and during the test. The sensing performance is compared to a DAS unit operating in the C-band on a parallel dedicated dark fiber in the same 250km submarine cable. The test was devised as part of the Center for Geophysical Forecasting (CGF) and the SUBmarine cableS for Research and Exploration (SUBMERSE) project, an innovative, EU-funded project which aims to utilize existing submarine cables, already used by the research and education networking

community, to monitor the Earth and its systems.

Our findings confirm the feasibility of both uncompromised sensing capability of L-band DAS, performing as well as C-band DAS, and uncompromised telecom performance in the same fiber. The concept was validated beforehand in lab experiments and on telecom links with synthetic traffic [4], and now, on a telecom link with live traffic.

Test Setup

Two unrepeaters submarine fiber optic cables of 250km run from Longyearbyen to Ny-Ålesund, outside Svalbard in the High Arctic. Each cable holds multiple optical fibers (G652.D). The map in Fig. 1(a) shows the cable location, where the blue area indicates where the cable is untrenched and located at shallow water depth (100 m). Between this area and Ny-Ålesund, the cable is located at much deeper water (250 m), in an unusually quiet area. In this test, two optical fibers in the innermost cable ("Indre Kabel", closest to the shore) are interrogated by two OptoDAS interrogators configured with a spatial resolution of 8 m. OptoDAS is based on frequency swept interrogation (FSI) developed by Alcatel Submarine Networks Norway AS (ASN). A C-band OptoDAS (1536.6 nm) was used to interrogate a dark fiber, and an L-band OptoDAS (1577.9 nm) was multiplexed using a standard C/L-filter onto a live telecom fiber running C-band channels. Both interrogated fibers reside in the same submarine cable structure. The DAS interrogators were located in Ny-Ålesund. The test was carried out using an in-production Nokia DWDM system with five active wavelengths connecting Ny-Ålesund and Longyearbyen in a distance of 250 km. Figure 1(b) shows a schematic of the test layout and the main active optical components involved in the transmission link. The dBm numbers in figure 1(b) indicate the power level of the SFM6 channels as they traverse through the system.

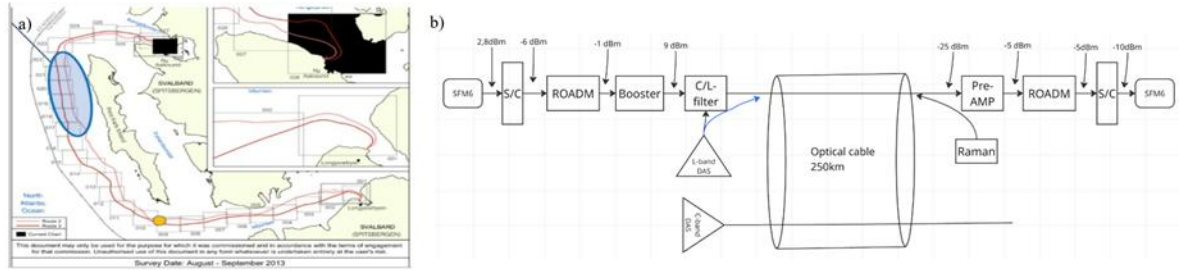


Figure 1: (a) Map showing “Ytre Kabel” (outer cable) and “Indre Kabel” (inner cable) from Longyearbyen to New Ålesund, Svalbard, Norway. C-band DAS, L-band DAS, and DWDM system are running over fibers in “Indre Kabel”. (b) Optical schematics of test.

Due to site specific constraints, the two DAS units had to be installed at a location different from the DWDM terminal. As a result, the fiber connecting the L-band DAS to the submarine cable goes through a chain of patch cords, a -3 dB splitter, 1 km of additional fiber, two Optical Distribution Frames (ODFs), and the previously mentioned C/L-filter before connecting to the submarine cable. The compound pathway accumulates a total one-way connection loss of 12 dB for the L-band DAS. The C-band DAS, which did not have to be merged with the DWDM terminal, was connected to the submarine cable through a more direct route, yielding only 2 dB of one-way connection loss. This 10 dB of imbalance in optical connection losses, corresponding to the loss of 50 km of fiber, complicated the comparison of the two DAS datasets. However, both DAS units were connected to a 10 km launch fiber spool to confirm that both DAS units produced the same data quality within these first 10 km. To validate uncompromised L-band DAS fidelity and telecom co-existence, we analyze and compare the recorded data from a regional earthquake with high SNR, which are expected to be the same given the identical design and performance of the C- and L-band OptoDAS units when both are deployed on dark fiber. To demonstrate that the telecom transmission is not affected by the L-band DAS, channel performance metrics such as pre-FEC BER, post-FEC BER and Q-margin are evaluated as these are the key performance

monitoring factors that indicate degradation on signal qualities [6].

Result

We evaluated the impact of coexistence from both the perspective of DAS and production telecom traffic.

DAS Performance: Fig. 2(a) shows the OptoDAS OTDR traces. The upper panel shows the trace of the C-band DAS on the dark fiber through the direct connection route with only 2 dB of one-way connection losses. The bottom panel in Fig. 2(a) shows the trace for the L-band DAS multiplexed onto the telecom link with the 12 dB one-way connection loss (due to the site-specific routing needed to connect the L-band DAS to DWDM terminal). The excess insertion loss of 10 dB is evident when comparing the two OTDR traces in Fig. 2(a). Note that in a permanent installation, the insertion loss should be minimized by splicing the connections, leaving only a 0.5 dB connection loss for the C/L-filter. The optical loss difference for the two DAS units corresponds with the difference in DAS noise floor characteristics seen in Fig. 2(b) showing waterfall RMS panels for the two DAS units.

Several catalogued seismic events were captured during our field-trial. Fig. 2(b) shows the C-band DAS (left) and L-band DAS (right) rms data in 3-25 Hz bandwidth for the magnitude 3.3 event occurring 2023-11-14T11:19:31 100 km away from the cable. The earthquake was clearly

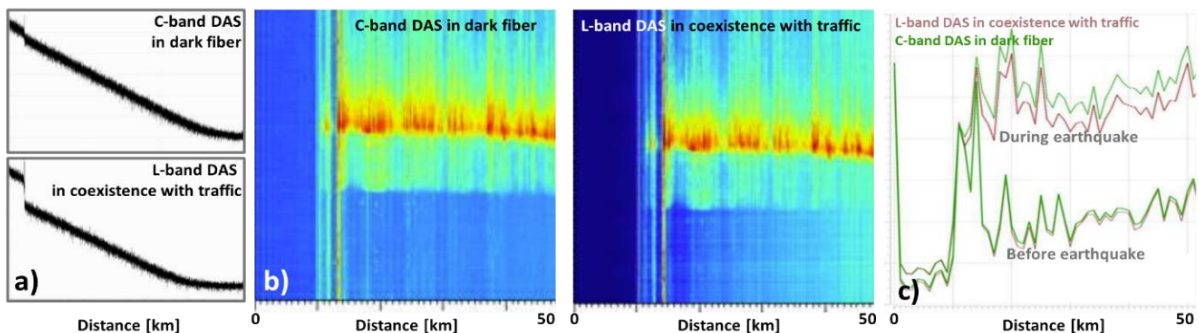


Figure 2: (a) OTDR traces, (b) rms waterfall, and (c) rms phase rate from a magnitude 3.3 event 100 km from cable.

visible in both datasets with similar amplitude. Fig. 2(c) compares the RMS phase rate vs. fiber length from 30 sec time windows at 400 m spatial averaging. The solid lines show that the average background response before the earthquake is the same. The dashed lines show the average while the earthquake is detected on the cable. Note that the top two lines from 0-10km are dashed lines. The amplitude on the L-band records on the lit fiber have the same distance dependence as the C-band DAS interrogating on dedicated dark fiber.

Performance of telecom wavelengths under coexistence with L-band DAS: We monitored five active telecom channel wavelengths with the fol-

addition, we compared measurements of simultaneous sensing data from a C-band DAS interrogating a dark fiber in the same submarine cable. Results show that the L-band DAS coexisting with traffic provides the same range and sensitivity as the C-band DAS on the dedicated dark fiber. Even though the L-band was connected to the submarine cable through a 10 dB of additional insertion loss, the analyzed events were adequately recorded on both the C-band DAS and L-band DAS. The results demonstrate that it is feasible for operators to perform fiber sensing even when dark fiber is not available by using the existing fibers of C-band DWDM channels by sensing in the L-band.

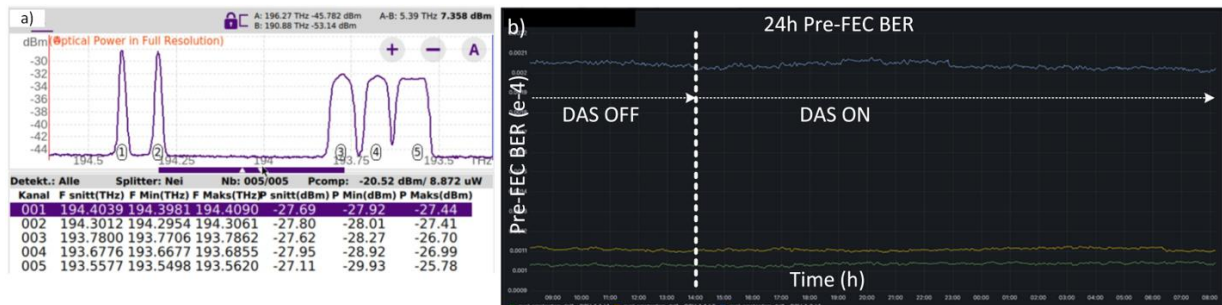


Figure 3: (a) Optical spectrum of running channels. (b) Pre-FEC bit error rate (BER) of three running channels during a 24-hour timeframe, L-band DAS switched on at 13:58 UTC.

lowing modulation and bitrates: 2 x 1G (OOK), 2 x 200G (62Gbaud QPSK with SDFEC-G2), 1 x 300G (84.23Gbaud 16SQAM with SDFEC-V). The 200G wavelengths carried 2 x 100G Ethernet traffic at 100% load with fixed frame length 1520 Bytes generated with a Viavi tester. The 300G wavelength carried production traffic. Fig. 3(a) shows the measurement through an optical spectrum analyzer monitoring the optical line for possible effects from the L-band.

The performance of the running DWDM traffic channels was monitored closely during a 14-day period when the L-Band DAS was multiplexed over the same optical fiber. The results confirm that there is no influence from DAS on the optical transmission line: no increase was detected in pre-FEC and post-FEC bit error rates, and no degradation of the Q-margin was observed. Fig. 3(b) shows the pre-FEC bit error rate (BER) of three C-band telecom channels (200G and 300G) during a 24-hour timeframe that includes the time when the DAS was turned ON at 13:58 UTC.

Conclusion

We have demonstrated uncompromised L-band DAS performance and telecom coexistence in a live telecom link carrying production traffic with DWDM signals in the C-band. No noticeable deterioration of pre-FEC BER, post-FEC BER or Q-margin of the data channels was observed. In

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